

HAT-P-20b

R-1364

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-20_170130 · created 2026-07-31T23:58:24+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2457783.72776 +/- 0.00096 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1068 +/- 0.0056
Transit depth (Rp/Rs)^2	1.14 +/- 0.12 [%]
Orbital Inclination (inc)	89.9 +/- 0.51
Airmass coefficient 1 (a1)	0.9632 +/- 0.0066
Airmass coefficient 2 (a2)	0.036 +/- 0.019
Scatter in the residuals of the lightcurve fit is	0.68 %
Best Comparison Star	None
Optimal Aperture	4.95
Optimal Annulus	6.6
Transit Duration (day)	0.0793 +/- 0.001

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1068 $\pm$ 0.0056 vs 0.1284 (deviation 3.21 σ)
Duration fit vs published	0.0793 d vs 0.0775 d (deviation 1.5 σ)
Mid-time O-C	2.4 min
Depth SNR	15.9
Residual scatter	0.68 %

Light curve

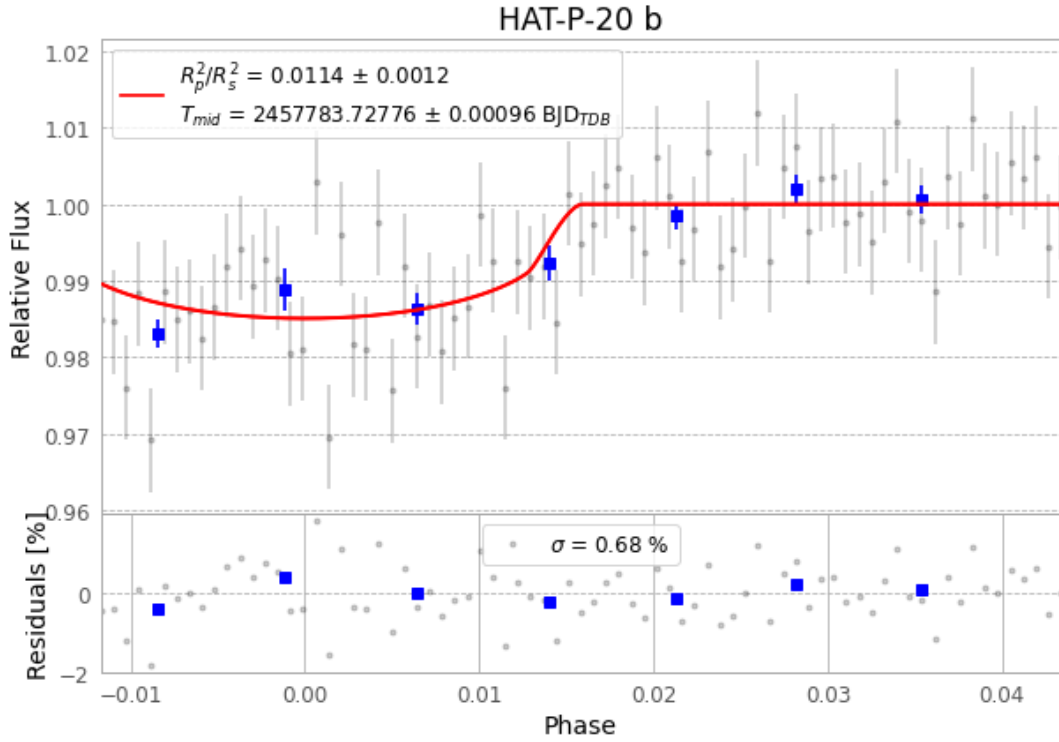


Figure 1 — FinalLightCurve_HAT-P-20 b_2017-01-29.png (SHA-256 48bbab8d0358b477...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [286, 279], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 ee44ff832b6f4108...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

77 FITS frames (51 MB), HATP-20170130043012.FITS → HATP-20170130081812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-20-170130.log (a250119c51f3...), FinalLightCurve_HAT-P-20 b_2017-01-29.png (48bbab8d0358...), FinalLightCurve_HAT-P-20 b_2017-01-29.csv (6e60751594c0...)

Compute resources

Wall time 140.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 23:58Z.